

STAT 153 & 248 - Time Series

Lecture Four

Fall 2026, UC Berkeley

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08 September 2026

We shall go over the main ideas behind frequentist and Bayesian inference in multiple linear regression.

1 Multiple Linear Regression

In multiple linear regression, we have one response variable y and m covariates $x_1, \dots, x_m$ ($m = 1$ corresponds to simple linear regression). We observe data on n instances or subjects for all these variables: $(y_i, x_{i1}, \dots, x_{im})$ for $i = 1, \dots, n$.

The following vector-matrix notation is very standard:

$$y = \begin{pmatrix} y_1 \\ \cdot \\ \cdot \\ \cdot \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1m} \\ 1 & x_{21} & \dots & x_{2m} \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ 1 & x_{n1} & \dots & x_{nm} \end{pmatrix}$$

Note the presence of the column of ones in the X -matrix. This notation is used not just to write formulae for linear regression, but also in code. For example, the OLS function in `statsmodels` uses the syntax `sm.OLS(y, X).fit()` to fit the linear regression model, where y ($n \times 1$ vector) and X ($n \times (m + 1)$ matrix) are defined above.

The multiple linear regression model (with normal errors) is given by:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_m x_{im} + \epsilon_i \quad \text{with } \epsilon_i \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2). \quad (1)$$

In the time series context, the model (1) arises in the following ways:

1. **Regression with functions of time:** Suppose we want to fit a quadratic function of time to the data. We can do this via the model (1) with x_1 being the time variable, and x_2 being the squared time i.e., $x_{i1} = i$ and $x_{i2} = i^2$. Suppose we want to fit a simple sinusoidal function to the data. We can do this via (1) with $x_{i1} = \cos(2\pi i/12)$ and $x_{i2} = \sin(2\pi i/12)$.
2. **AutoRegression (AR):** If we take $x_{ij} = y_{i-j}$, then we get the AR model:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_m y_{t-m} + \epsilon_t.$$

The idea here is that we are using the m most recent values of the time series to predict the next observation. AR models are very commonly used and they work quite well for time series.

Let us focus for now on regression using functions of time. We shall study AutoRegression in detail later.

Our goal is to use the observed data (in y, X) to obtain estimates along with uncertainty intervals for the parameters $\beta_0, \dots, \beta_m$. There are two broad principles for doing this: frequentist and Bayesian.

2 Frequentist Inference for Linear Regression

The basic ideas behind frequentist inference are as follows:

1. Construct a method for estimating the unknown parameters. In the simple linear regression context, one can use least squares to estimate $\beta_0, \beta_1, \dots, \beta_m$. Probably the most popular estimation strategy is Maximum Likelihood Estimation (this requires writing down a likelihood function).
2. Calculate (exactly or using some approximations) the distribution of the estimators. Use quantiles of the distribution for obtaining interval estimates for the unknown parameters. The quantiles might themselves depend on other unknown parameters (which would then have to be replaced by estimates).

2.1 Estimates

2.1.1 Least Squares Estimates

For linear regression, the estimates are obtained by the method of least squares, where the sum of squares

$$S(\beta_0, \beta_1, \dots, \beta_m) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2 \quad (2)$$

is minimized over all values of $\beta_0, \beta_1, \dots, \beta_m$. The minimizing values $\hat{\beta}_0, \dots, \hat{\beta}_m$ are known as least squares estimates.

Using the notation:

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1m} \\ 1 & x_{21} & \dots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nm} \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix} \quad \hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_m \end{pmatrix},$$

the sum of squares function $S(\beta_0, \dots, \beta_m)$ can be written as

$$S(\beta) = \|y - X\beta\|^2.$$

The least squares estimator $\hat{\beta}$ is given by the formula:

$$\hat{\beta} = (X^T X)^{-1} X^T y. \quad (3)$$

The proof of (3) is as follows. The gradient of $S(\beta)$ is given by

$$\begin{aligned} \nabla S(\beta) &= \nabla [\|y - X\beta\|^2] \\ &= \nabla [(y - X\beta)^T (y - X\beta)] \\ &= \nabla [y^T y - \beta^T X^T y - y^T X \beta + \beta^T X^T X \beta] = 2X^T y - 2X^T X \beta. \end{aligned}$$

Because $\hat{\beta}$ minimizes $S(\beta)$, the gradient should equal zero when $\beta = \hat{\beta}$, and this leads to

$$X^T (y - X\hat{\beta}) = 0 \implies X^T X \hat{\beta} = X^T y \implies \hat{\beta} = (X^T X)^{-1} X^T y. \quad (4)$$

2.1.2 Maximum Likelihood Estimates (MLEs)

Under the assumption $\epsilon_i \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$, we can write the likelihood explicitly and maximize it to obtain MLEs. As seen below, the MLE of $\beta_0, \dots, \beta_m$ will coincide with least squares, but the ML method additionally will give an estimate of σ .

With the normality assumption $\epsilon_i \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$, the linear regression model can be rewritten as:

$$y_i \stackrel{\text{independent}}{\sim} N(\beta_0 + \beta_1 x_{i1} + \dots + \beta_m x_{im}, \sigma^2). \quad (5)$$

The likelihood then becomes:

$$\begin{aligned} f_{y_1, \dots, y_n | \beta_0, \dots, \beta_m, \sigma}(y_1, \dots, y_n) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2}{2\sigma^2}\right) \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2\right) \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) \end{aligned} \quad (6)$$

Recall $S(\beta_0, \dots, \beta_m)$ above is the sum of squares defined in (2). To write this likelihood, we are assuming that $x_1, \dots, x_n$ are fixed. This assumption is fine if $x_i = i$ (regression with time as covariate) but not strictly true when $x_i = y_{i-1}$ (auto-regression). We shall see how it is still approximately true in the case of AutoRegression later.

Another way to write the likelihood is to note that (5) is equivalent to:

$$y \sim N_n(X\beta, \sigma^2 I_n). \quad (7)$$

In other words, the n -dimensional vector y is multivariate normal with mean $X\beta$ and covariance $\sigma^2 I_n$. Recall that the density of the multivariate normal $y \sim N_n(\mu, \Sigma)$ is given by:

$$\frac{1}{(2\pi)^{n/2}} \frac{1}{\sqrt{\det \Sigma}} \exp\left(-\frac{1}{2}(y - \mu)^T \Sigma^{-1} (y - \mu)\right).$$

Thus the density corresponding to (7) is:

$$(2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{1}{2}(y - X\beta)^T (\sigma^2 I_n)^{-1} (y - X\beta)\right) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{\|y - X\beta\|^2}{2\sigma^2}\right)$$

which is the same as (6) because $S(\beta) = \|y - X\beta\|^2$.

The Maximum Likelihood Estimates of the parameters β, σ are obtained by maximizing the likelihood. As maximizing a function is equivalent to maximizing its logarithm, we attempt to maximize the log-likelihood which leads to an easier maximization. The log-likelihood is:

$$\begin{aligned}\log\text{-likelihood} &= \log \left[(2\pi)^{-n/2} \sigma^{-n} \exp \left(-\frac{S(\beta)}{2\sigma^2} \right) \right] \\ &= -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{S(\beta)}{2\sigma^2}.\end{aligned}$$

To maximize the log-likelihood, we simply take derivatives with respect to the unknown parameters β, σ and equate those to zero:

$$\begin{aligned}\nabla_{\beta} \log\text{-likelihood} &= -\frac{1}{2\sigma^2} \nabla_{\beta} S(\beta) = 0 \implies \nabla_{\beta} S(\beta) = 0 \\ \frac{\partial}{\partial \sigma} \log\text{-likelihood} &= -\frac{n}{\sigma} + \frac{S(\beta)}{\sigma^3} = 0 \implies \sigma = \sqrt{\frac{S(\beta)}{n}}.\end{aligned}$$

The first equation coincide with the corresponding equations (4) for minimizing least squares. This shows that the MLE for β coincides with the least squares estimators (3). The MLE for σ is given by the third equation above (with β replaced by $\hat{\beta}$ respectively):

$$\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{S(\hat{\beta})}{n}}. \quad (8)$$

To summarize, under the assumption of normality on the errors $\epsilon_i \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2)$, the MLE of β coincides with the least squares estimator (3), and the MLE of σ is given by (8).

2.2 Distribution of Estimates

Here we need to calculate the distribution of the estimates. For $\hat{\beta}$ (given by (3)), using the fact (7) we can write

$$\hat{\beta} = (X^T X)^{-1} X^T y \sim N((X^T X)^{-1} X^T X \beta, \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1}) = N(\beta, \sigma^2 (X^T X)^{-1}).$$

Here we used the following fact:

$$X \sim N(\mu, \Sigma) \implies AX \sim N(A\mu, A\Sigma A^T).$$

Because $\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$, it is clear that $\hat{\beta}$ is unbiased for β .

The distribution of $\hat{\sigma}_{\text{MLE}}$ is given by:

$$\frac{n\hat{\sigma}_{\text{MLE}}^2}{\sigma^2} \sim \chi_{n-m-1}^2$$

where χ_{n-m-1}^2 is the chi-squared distribution with $n - m - 1$ degrees of freedom. We will not go into the proof of this fact.

The mean of the chi-squared distribution equals its degrees of freedom which implies that

$$\mathbb{E} \hat{\sigma}_{\text{MLE}}^2 = \sigma^2 \frac{n - m - 1}{n}.$$

Therefore the MLE for σ^2 is not unbiased (in contrast, the MLEs $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased). It is easy to correct the bias leading to the following unbiased estimator of σ^2 :

$$\hat{\sigma}_{\text{unbiased}}^2 = \frac{n}{n-m-1} \hat{\sigma}_{\text{MLE}}^2 = \frac{S(\hat{\beta})}{n-m-1}.$$

Usage of $\hat{\sigma}_{\text{unbiased}}$ is much more common than that of $\hat{\sigma}_{\text{MLE}}$ (note that $\hat{\sigma}_{\text{unbiased}}$ is not unbiased for σ ; rather the square of $\hat{\sigma}_{\text{unbiased}}$ is unbiased for σ^2).

Another fact is that $\hat{\beta}$ and $\hat{\sigma}_{\text{unbiased}}^2$ are independent.

These facts are used to derive the following confidence interval for a component β_j of β :

$$\left[\hat{\beta}_j - \hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}} t_{n-m-1, \alpha/2}, \hat{\beta}_j + \hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}} t_{n-m-1, \alpha/2} \right] \quad (9)$$

where $t_{n-m-1, \alpha/2}$ is the positive point such that $\mathbb{P}\{t_{n-m-1} \geq t_{n-m-1, \alpha/2}\} = \alpha/2$ (i.e., the t -distribution with $n-m-1$ degrees of freedom assigns probability mass $\alpha/2$ to the right of $t_{n-m-1, \alpha/2}$). Also $(X^T X)^{j+1, j+1}$ is the $(j+1, j+1)$ -th diagonal entry of $(X^T X)^{-1}$. Note that $\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$ so that $\hat{\beta}_j \sim N(\beta_j, \sigma^2 (X^T X)^{j+1, j+1})$ (the index is $j+1$ instead of j on the right hand side because the first component of $\hat{\beta}$ is $\hat{\beta}_0$ and not $\hat{\beta}_1$).

(9) is a valid confidence interval because:

$$\frac{\hat{\beta}_j - \beta_j}{\sigma (X^T X)^{j+1, j+1}} \sim N(0, 1) \text{ and } \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} (X^T X)^{j+1, j+1}} \sim t_{n-m-1}$$

where t_{n-m-1} is the t -distribution with $n-m-1$ degrees of freedom.

This section on deriving the distributions of the frequentist estimates has been discussed for completeness. We will not be using these facts or ideas in this course.

3 Bayesian Inference for Linear Regression

In Bayesian inference for (1), we work with the prior

$$\beta_0, \beta_1, \dots, \beta_m, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{unif}(-C, C)$$

for a very large positive C . The joint posterior density of $\beta_0, \dots, \beta_m, \sigma$ is then given by

$$\begin{aligned} f_{\beta_0, \beta_1, \dots, \beta_m, \sigma | \text{data}}(\beta_0, \beta_1, \dots, \beta_m, \sigma) \\ \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) I\{-C < \beta_0, \beta_1, \dots, \beta_m, \log \sigma < C\}. \end{aligned}$$

where we use the notation

$$S(\beta_0, \beta_1, \dots, \beta_m) := \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2$$

for the sum of squares.

The posterior over only the coefficient parameters β_0, β_1 can be obtained by integrating (or marginalizing) the parameter σ .

$$\begin{aligned}
& f_{\beta_0, \beta_1, \dots, \beta_m | \text{data}}(\beta_0, \beta_1, \dots, \beta_m) \\
&= \int f_{\beta_0, \beta_1, \dots, \beta_m, \sigma | \text{data}}(\beta_0, \beta_1, \dots, \beta_m, \sigma) d\sigma \\
&\propto I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \int_{e^{-C}}^{e^C} \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) d\sigma \\
&\approx I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) d\sigma \\
&= I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \left(\frac{1}{S(\beta_0, \dots, \beta_m)}\right)^{n/2} \int_0^\infty s^{-n-1} \exp\left(-\frac{1}{2s^2}\right) ds \\
&\propto I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \left(\frac{1}{S(\beta_0, \dots, \beta_m)}\right)^{n/2} \\
&\approx \left(\frac{1}{S(\beta_0, \dots, \beta_m)}\right)^{n/2} \propto \left(\frac{S(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)}{S(\beta_0, \beta_1, \dots, \beta_m)}\right)^{n/2}
\end{aligned}$$

where $\hat{\beta}_0, \dots, \hat{\beta}_m$ denote the least squares estimators of $\beta_0, \dots, \beta_m$ (i.e., $(\hat{\beta}_0, \dots, \hat{\beta}_m)$ minimizes $S(\beta_0, \dots, \beta_m)$ over all values of $\beta_0, \dots, \beta_m$).

Our posterior density for $\beta_0, \dots, \beta_m$ is thus:

$$f_{\beta_0, \beta_1, \dots, \beta_m | \text{data}}(\beta_0, \beta_1, \dots, \beta_m) \propto \left(\frac{S(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)}{S(\beta_0, \beta_1, \dots, \beta_m)}\right)^{n/2}. \quad (10)$$

In the next lecture, we will explain why this is a multivariate t -density.